Where Pedestrians Die, and Why

Explainable machine learning on 60,000 pedestrian fatalities, translated into plain-language, map-based decision support

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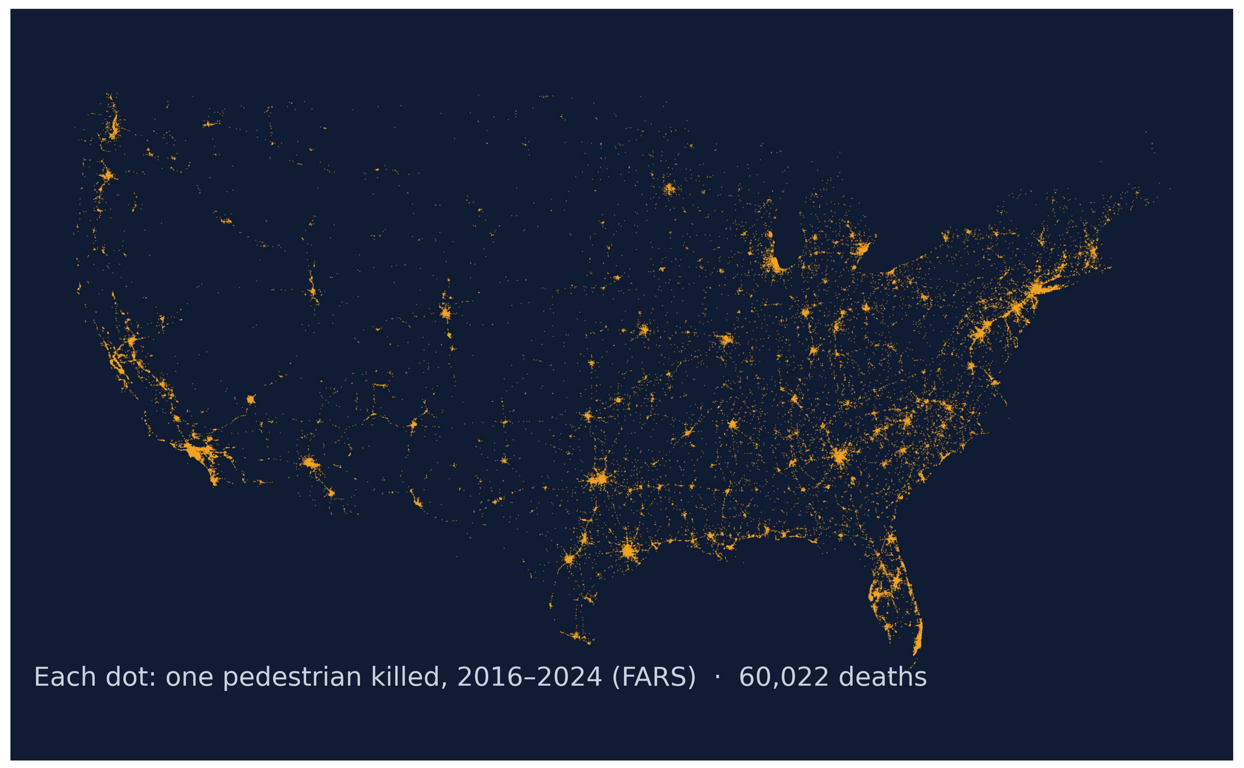
60,022
pedestrian deaths, mapped and explained

1 in 3
died in one scenario: dark urban arterials, midblock

19.4%
of fatal crashes involve a pedestrian, up from 17.4%

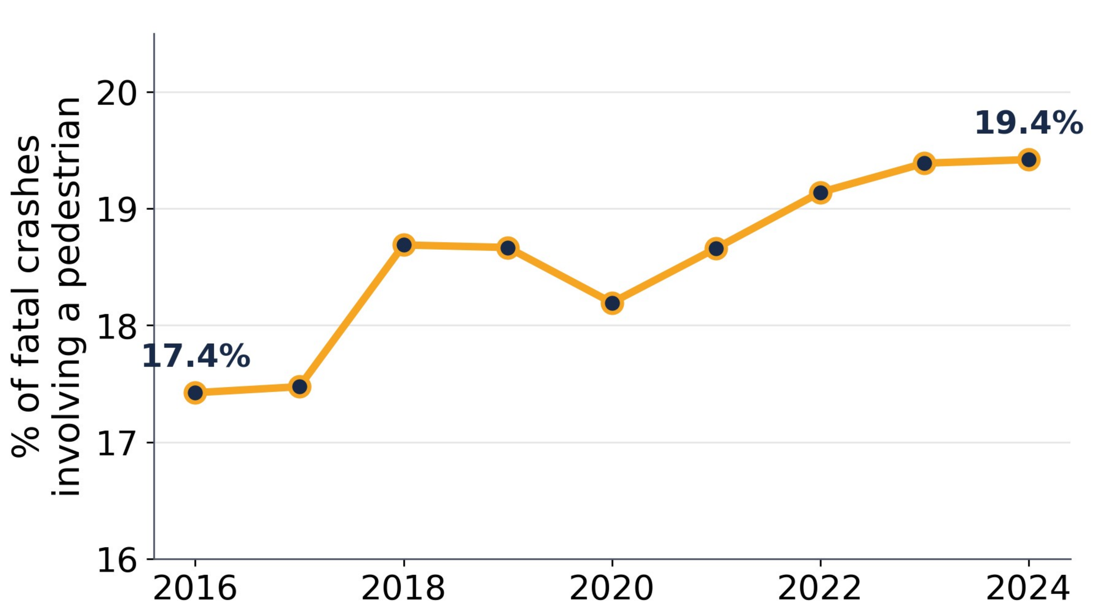
1 Introduction & Data



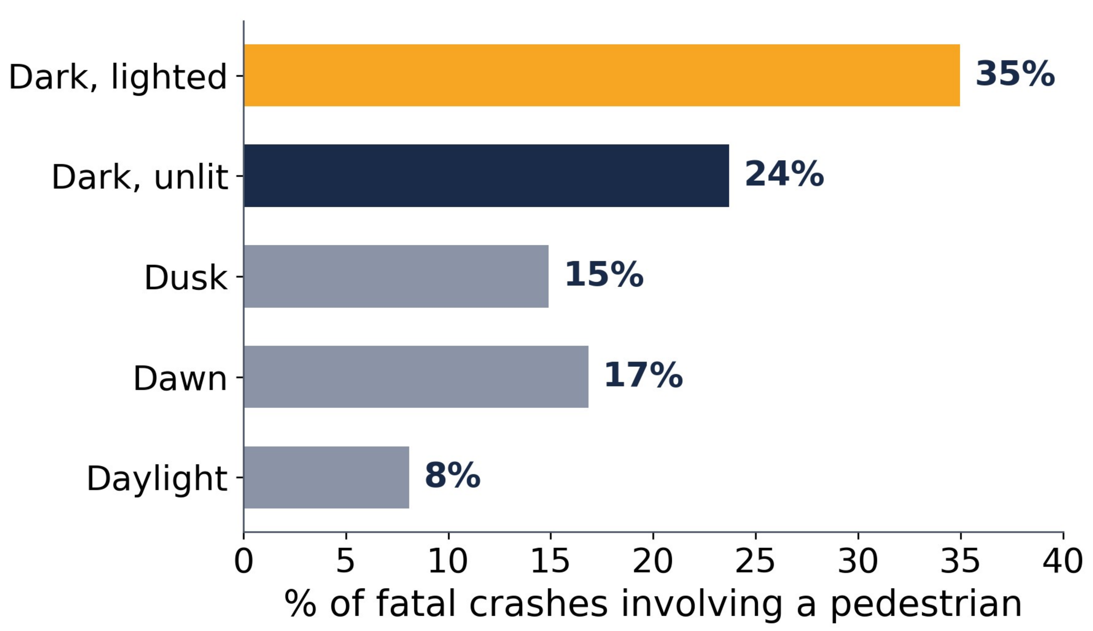
Pedestrians are the fastest-growing share of U.S. road deaths. From NHTSA's Fatality Analysis Reporting System (FARS), the census of every fatal crash in America, we assembled 2016–2024: 325,922 fatal crashes, 60,562 involving a pedestrian, joined with county-level Census ACS context.

Research question: given that a fatal crash occurred, what distinguishes the ones that kill people on foot? And can every one be explained?

Pedestrian share of fatal crashes keeps climbing



The dark-hours gradient



2 Methods

FARS + ACS: five files per year + county context

Crash-level dataset: 1 row per fatal crash · 18.6% pedestrian

Feature discipline: pre-crash information only: 16 variables

XGBoost: stratified 80/20 · class-weighted 4.4:1

SHAP: exact TreeSHAP · top-3 factors per crash

Narratives: who, where, struck by what, why + fix

Interactive map: 60,022 explained dots · any phone

Why this target is valid. FARS holds only fatal crashes, so fatal-vs-nonfatal cannot be modeled from it, but pedestrian-involved vs not can: both classes exist. Post-outcome variables (first harmful event, collision manner, hit-and-run) are excluded as leakage.

Validation (held-out 65,185 crashes)

Model	AUC	PR-AUC	Recall
Logistic (baseline)	0.869	0.616	0.82
XGBoost (final)	**0.885**	**0.658**	**0.81**
Train 16–22 → test 23–24	0.888	0.667	0.82

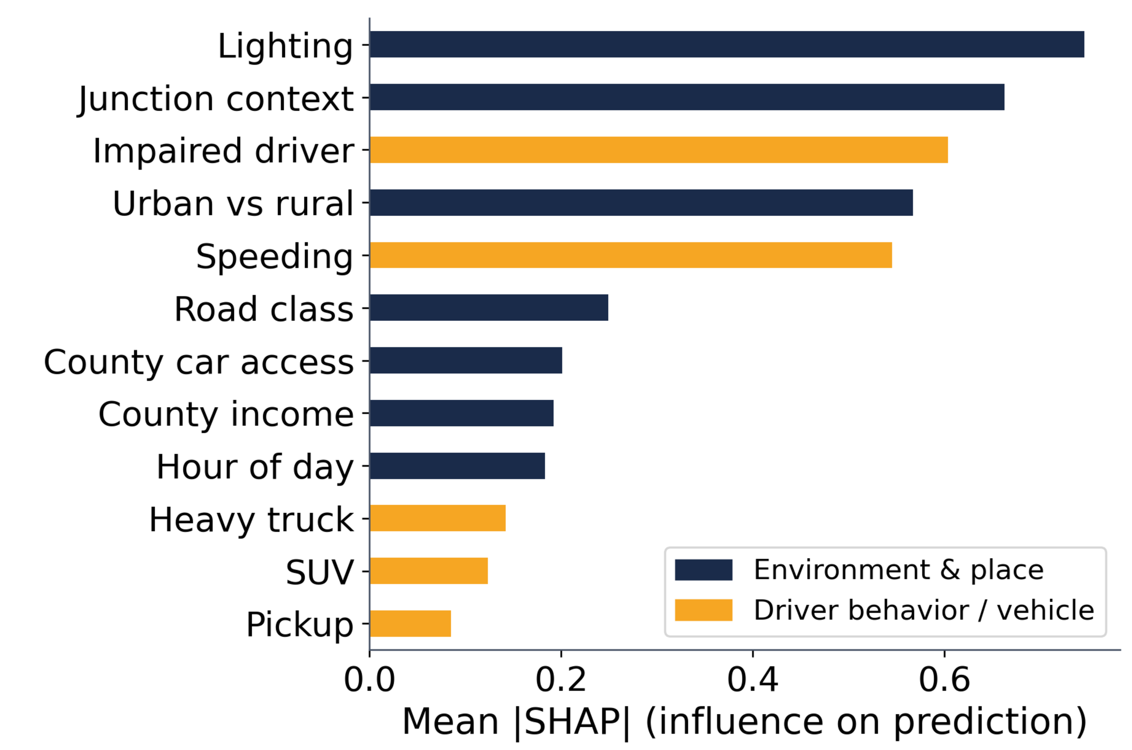
Performance holds on unseen years: stable, not memorized. Precision 0.49 by design: screening accepts false alarms.

The data at a glance

325,922 fatal crashes	**60,562** pedestrian deaths
99.3% mappable	**3,144** counties (ACS)
16 → 48 features encoded	**60,022** dots on the map

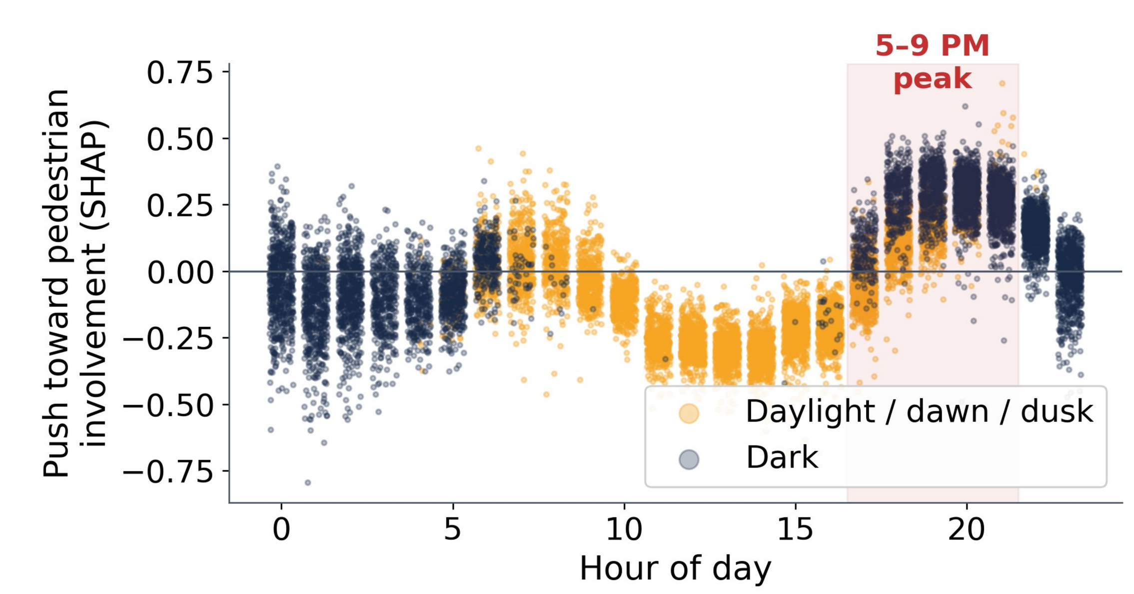
3 Findings: What Separates Them

The road AND the driver, in opposite directions



Impairment and speeding rank high but push AWAY from pedestrian involvement: those crashes kill vehicle occupants. Pedestrian deaths are an environment story.

Risk peaks 5–9 PM: the evening dark



Context interactions (conditional SHAP)

- Dark-unlit roads matter 2–3× more on arterials and freeways than on local roads
- The arterial effect roughly doubles after dark (0.19 vs 0.08)
- Evening hours carry ~4× more pedestrian risk when dark (0.26 vs 0.06)

Descriptive epidemiology. 74% of pedestrian deaths occur midblock. 64% of hit-and-run fatal crashes kill a pedestrian (vs 15% otherwise). Median victim age 48; one in five is 65+; 4% are children. SUV/crossover share rose 19% → 25%.

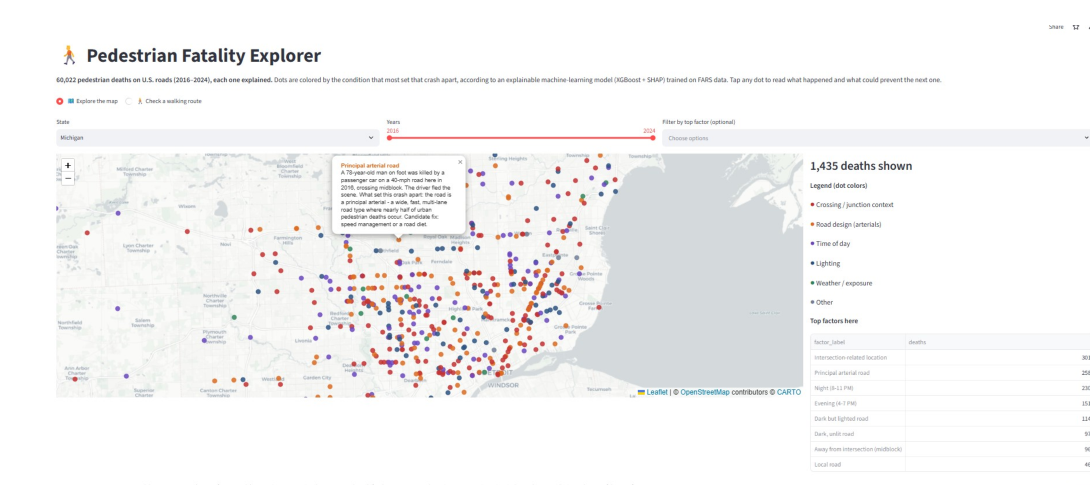
4 From Evidence to Practice

One scenario dominates, and it is countermeasure-shaped

Crash profile	% of deaths	Ped share	Lift
Dark	75.7%	28.7%	1.5×
+ Urban	62.9%	36.6%	2.0×
+ Arterial road	42.5%	42.2%	2.3×
+ Away from intersection	**29.5%**	**45.5%**	**2.5×**

- **Dark / unlit** → continuous, crossing-focused lighting
- **Midblock deaths** → hybrid beacons, median refuges
- **Arterials** → road diets, speed management
- **Intersections** → leading pedestrian intervals

Every dot, explained. On your phone.



Scan to open the live map. Tap any dot: who was killed, where, struck by what, why, and a candidate countermeasure.

*"A 22-year-old man was killed by a pickup truck on a 70-mph road, crossing midblock in the dark, with no street lighting. **Candidate fix: lighting and evening speed enforcement.**"*

Takeaways

1. Risk is concentrated, not diffuse: one scenario holds 1 in 3 deaths, and every element has a proven fix.

2. The deadliest window is the evening dark: target lighting and speed where evening foot traffic is.

3. Explainability is the bridge from classifier to corridor-level decision support anyone can read.

Read honestly. Associations, not causes. No exposure denominator in fatal-only data; no county-level equity gradient (tract-level: future work). Pipeline + app: github.com/ErfanRamezanpour99/ped-fatality-explorer